

Sri Vallabha Deevi

DIRECTOR, DATA SCIENCE, TIGER ANALYTICS | ADJUNCT FACULTY, DEPT. OF DS & AI, IIT MADRAS

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Experience

Tiger Analytics

Chennai, India

DIRECTOR - DATA SCIENCE

Jan 2018 - present

- Responsibilities - Technical solutioning of projects, **Applied Research** to evaluate applicability of new algorithms for business problems, Team building - Guiding teams on project delivery, Mentoring teammates on Learning & Growth, Client interactions for Business Development.
- Full truck load **freight cost prediction** for a US Third Party Logistics company. **Linear Regression** model with feature engineering to predict prices accurately on different routes with varying amount of history. Solution in use since 2020 and improved margins and conversion rates.
- Digital offer recommendation using **Reinforcement Learning** for the largest Quick Service Restaurant chain in the world. Offer recommendation is treated as action selection problem for customer in a state. State is a set of features derived from recent transactions of the customer. Trained a **Deep Q Network** in TensorFlow to predict best actions in a given state to improve quarterly customer revenue.
- Accelerating design verification for a large semiconductor memory company using **Generative Adversarial Networks** and **Graph Neural Networks**. Using Machine Learning to predict the outcomes of bench marking test results, based on historical data of chip design and test performance. Development of ML Models to design test benches that accelerate code coverage for Finite State Machines.
- Route optimization** for delivery of new cars from yards to dealers, for a US automotive consortium. Variant of **Capacitated Vehicle Routing Problem** with solution balancing truck packing, vehicle routing and inventory holding time constraints. Problem solved multiple times in a day, as new inventory reaches the yards.
- Fuel efficiency** prediction based on truck driving behaviour for the largest commercial vehicle manufacturer in India. **Regression** models with feature engineering to predict mileage for different trucks on different terrains. Prescriptions on driving attributes to improve fuel efficiency based on observations from other drivers on that route.
- Industrial safety compliance monitoring using **Convolutional Neural Networks**. Fine tuned baseline object detection models to detect hard hats and other PPEs. Relating person & PPE detection to identify safety violations. Zoning implementation to detect unauthorized entries.
- Generative AI** based solution to improve business process for a large Electronic Device Manufacturer. **Retrieval-Augmented Generation** to answer questions from policy documents. **Code generation** based on questions to retrieve data from data bases and summarize findings. Integration with client internal Chatbot to scale across organization.
- Marketing channel efficiency measurement using **Multi Touch Attribution** and **Market Mix Modeling** for a US beverage retailer. Using channel wise advertising impressions to predict sales, accounting for memory and saturation effects of advertisements. Channel ROI measurement and spend optimization.
- Recommendation System** for email marketing and **Propensity Modeling** for in-store coupon delivery for the same retailer. **Cosine Similarity** and **Hybrid Recommender** used to recommended new products to customers based on recent purchases. Model evaluated by client against existing method, and currently being used to make monthly recommendations since 2020.
- Equipment failure prediction via **Anomaly Detection** using IoT sensor data for the largest steel company in India. For steady load operations, **Multi-variate Gaussian** process based model to identify the proportion of outliers with time. For variable load operations, **Regression** model based errors to predict anomalies.
- Process control using **Reinforcement Learning** for optimal product properties, for the same steel company. Process input parameters constitute the state, controllable parameters form the action space and Tensile strength of steel is the output parameter. Objective function is designed to limit Tensile strength to a specific range. Trained a **Q learning** based model, with tabulated states and actions.
- Hypothesis testing** to relate sales patterns and margins for a national beverage distributor in the US. Using statistical measures on observed variables to validate hypothesis. Provided a quantitative approach for the client to verify their beliefs.
- AB testing** of product features, conversion propensity modeling for an anti-virus software company. Features introduced by product team are tested on a subset of customers, to assess their impact on critical business outcomes like engagement and renewal rates. Estimating required sample sizes to minimize probability of Type 1 errors and maximize the **Power of Test**.
- Attribution modeling - in store product placement effectiveness, production activities driving energy consumption. **Linear Regression** models to relate outcome to input variables, to estimate the impact of each input variable on the outcome. Simple models to enable **Explainability**.
- Warehouse inventory management using **Reinforcement Learning** for a US consumer food company. State consists of current inventory, incoming supplies, outgoing orders of that week. Actions are discrete order quantities. Reward is the overall operating cost, including storage cost, unfulfilled order cost and product wastage cost.

Indian Institute of Technology Madras

Chennai, India

SENIOR SCIENTIST

Aug 2015 - Dec 2017

- Worked in Multiphase Flow Physics Laboratory, Dept. of Applied Mechanics with **Prof. Mahesh V. Panchagnula**.
- Statistical analysis of experimental data from sprays - **Time Series Modeling, Extreme Value Analysis, Conditional Probability**.
- Modeling breakup mechanisms of liquid droplets using Population Balance Model and Euler-Euler formulation of two phase flow.
- Mathematical Modeling and Simulation of active matter. Statistical studies of surface properties based on deposited droplet behaviour.

Emmeskay Systems

ENGINEER

Chennai, India

Apr 2008 - Jun 2009

- Mathematical modeling of automotive systems for concept evaluation and **Control Strategy** development.
- Design analysis of **Dual Mass Flywheel & Hybrid Electric Vehicle** drive train in various driving cycles.
- Design of **Hybrid Electric Tricycle** with manual and electric propulsion.

Deloitte Consulting

BUSINESS ANALYST

Hyderabad, India

Jul 2007 - Mar 2008

- Primary surveys to identify world wide suppliers for personal protective equipment.
- Model to improve sourcing options for 1500 SKUs across nine geographies for a safety consumables company in the US.
- Data analysis to de-duplicate customers of an Aviation Maintenance, Repair and Overhaul company. Developed Excel Macros to process duplicates and clean the data.

Education

Indian Institute of Science

DOCTOR OF PHILOSOPHY IN AEROSPACE ENGINEERING

Bangalore, India

Jul 2009 - Jul 2015

- Worked in Laboratory for Turbulence Computations, Dept. of Aerospace Engineering with **Prof. Joseph Mathew**.
- Computational study of fuel spray evaporation in gas turbine engine using **High Performance Scientific Computing**.
- Developed a 3D **Parallel Computing** code in **FORTRAN** to calculate properties of continuous phase, and Lagrangian tracking for droplets.
- Established the usage of **Explicit Filtering Large Eddy Simulation** for predicting dilute multi phase flows.

Indian Institute of Technology Madras

BACHELOR OF TECHNOLOGY IN MECHANICAL ENGINEERING, CGPA: 9.14/10

Chennai, India

Jul 2003 - Jun 2007

- B.Tech. project on CFD study of Reverse Flow Gas Turbine Combustor at IC Engines Laboratory with **Prof. V. Ganesan**.
- Finished in **top 5** in a class of **110**, with focus on Thermal Engineering and Minor in **Marketing Management**.
- Summer internship on **Maintenance Management** at Hazira Petrochemical Complex, Reliance Industries.
- Andhra Pradesh Chief Minister's **Pratibha Award Gold Medal** for outstanding performance (top 0.4%) in IIT-JEE 2003.

Publications

Journals

- **Sri Vallabha Deevi**, Nachiketa Janardan, Mahesh V. Panchagnula. Shapes of Splattered Drops. Langmuir. Vol 33(18). pg 4592-4600. Apr 2017.
- Akshay S. Acharya, **Sri Vallabha Deevi**, Dhivyaraja Kumaran, Arun K. Tangirala, Mahesh V. Panchagnula. Spatio-temporal microstructure of sprays: data science-based analysis and modelling. Journal of Fluid Mechanics. Vol 912. A19. Feb 2021.

Conferences

- **Sri Vallabha Deevi**, Joseph Mathew. Large Eddy Simulations of Evaporating Spray Jets. ASME Turbo Expo 2015, Montreal, Canada. Aug 2015.
- **Sri Vallabha Deevi**, Joseph Mathew. Large Eddy Simulation of Particle Laden Flows. ILASS-Asia 2016, Chennai, India. Nov 2016.
- Dhivyaraja Kumaran, **Sri Vallabha Deevi**, Mahesh V. Panchagnula. Drop Size Distribution Effects in Intersecting Sprays. ILASS-Asia 2016, Chennai, India. Nov 2016.
- Bhaumik Shah, **Sri Vallabha Deevi**, Mahesh V. Panchagnula, Arun K. Tangirala. Identifying Non-Markov Effects in Sprays. ILASS-Asia 2016, Chennai, India. Nov 2016.
- Nachiketa Janardan, **Sri Vallabha Deevi**, Mahesh V. Panchagnula. Shapes of Randomly Placed Drops. APS DFD 2016, Portland, USA. Nov 2016.
- Dhivyaraja Kumaran, **Sri Vallabha Deevi**, Mahesh V. Panchagnula, Arun K. Tangirala. Non-Markov Effects in Intersecting Sprays. APS DFD 2016, Portland, USA. Nov 2016.

Thesis Guidance

Indian Institute of Technology Madras

CO GUIDE, INDUSTRIAL MATHEMATICS AND SCIENTIFIC COMPUTING, DEPT. OF MATHEMATICS

Chennai, India

2019 - present

- Jalua, Tumpa. 2024. Identification of strategies in multiplayer games using Reinforcement Learning. Guide - Dr. Sivaram Ambikasaran.
- Sreedevi, M. 2024. Advancing Design Verification: Leveraging ML for Enhanced Validation Process. Guide - Dr. Sivaram Ambikasaran.
- Sing, Sandeep. 2023. Customer Behaviour Simulator for training RL agents: Real world application. Guide - Prof. S. Sundar.
- Tyagi, Deepanshu. 2023. A survey of different recommendation algorithms and their applications. Guide - Prof. S. Sundar.
- Bag, Doyel. 2023. Applications of Machine Learning in Stock Market. Guide - Prof. S. Sundar.
- Sonker, Kavita. 2023. Explainable Artificial Intelligence. Guide - Prof. S. Sundar.
- Karmakar, Shreya. 2020. Sales impact of interaction between advertising channels or Market Mix Model. Guide - Prof. S. Sundar.

Teaching

Invited Talks

- Predictive to Prescriptive Analytics, Master Class. KJ Somaiya Institute of Management, Mumbai. Feb 2024.
- Data Science and Machine Learning for Engineers. **IIT Tirupati**, Tirupati. Nov 2023.
- Generative AI and Large Language Models. VIT, Chennai. Nov 2023.
- AI & ML: Introduction & Applications in Processing and Fabrication of Advanced Materials. PFAM 29, **IIT Tirupati**, Tirupati. Sep 2023.
- High Performance Computing & AI for Pharma Research. KL University, Vijayawada. May 2023.
- Explainable AI. Loyola Institute of Business Administration, Chennai. Dec 2022.
- AIML, Applications in Industry. **NIT Silchar**, Silchar. Mar 2022.
- Reinforcement Learning & Digital Marketing. GH Raisonni Institute of Engineering and Technology, Nagpur. Jan 2022.
- Machine Health Monitoring with AI. **ODSC APAC**, Virtual. Dec 2020.
- Why AI? Applications from the Real World. **Cypher 2019**, Bangalore. Sep 2019.
- Automating Industrial Safety Monitoring using Video Analytics. **Data Hack Summit 2018**, Bangalore. Nov 2018.

Workshops & Faculty Development Programs

- Exploring Explainable and Generative AI in Contemporary ML. Adi Sankara Institute of Engineering and Technology, Kalady. Jan 2024.
- Generative AI and Natural Language Processing. GH Raisonni College of Engineering, Nagpur. Jan 2024.
- Digital Manufacturing and Automation in Foundry. **IIT Tirupati**, Tirupati. Oct 2023.
- Predictive & Prescriptive Analytics. KJ Somaiya Institute of Management, Mumbai. Mar 2023.
- Trending Technologies in Computer Science. Dr. MGR University, Chennai. Oct 2021.
- Reinforcement Learning. **NIT Warangal**, Warangal. Jul 2021.
- Applied Data Science. BMS College of Engineering, Bangalore. Jul 2020.

Courses

- Taught Compressible Flows, as a part of **AM2530 : Foundations of Fluid Mechanics** offered by Prof. Mahesh V. Panchagnula, Dept. of Applied Mechanics, IIT Madras. 2016.
- Physics Faculty for Foundation Science and Technology program at **BASE** Bangalore. Weekend sessions for students of **Classes IX and X**, at various centers in Bangalore. 2010 - 2014.

Skills

Languages	Python, R, SQL, PySpark, CUDA, C, C++, FORTRAN
Engineering	MATLAB, OpenFOAM, Scilab, Octave, Fluent, AutoCAD, ParaView
Assorted	TensorFlow, PyTorch, Linux, git, LaTeX, CFD, Statistics, Robotics
Learning	Data Camp profile: https://www.datacamp.com/profile/srivallabhad

Accomplishments

Industry Expert, Board of Studies

- GH Raisonni College of Engineering, Nagpur. 2018 onwards.
- Loyola Institute of Business Administration, Chennai. 2020 onwards.
- Amrita Vishwa Vidyapeetham, Coimbatore. 2023 onwards.
- Anna University, Chennai. 2021.
- Galgotias University, Noida. 2020.
- SRM University, Chennai. 2019.
- Vellore Institute of Technology, Chennai. 2018.

Positions of Responsibility

- Panelist, **Reliance Foundation Post Graduate Scholarships**. 2021 onwards.
- Reviewer, Advanced Network Technologies and Intelligent Computing, Institute of Science, **Benaras Hindu University**. 2022 onwards.
- Reviewer, International Conference on Recent Trends in Microelectronics, Automation, Computing & Communication Systems, IEEE. 2022.
- Evaluator, **Smart India Hackathon**, Software & Hardware Editions. 2020.

Certifications

- Machine Learning by Andrew NG, Coursera.
- Machine Learning Engineering for Production by DeepLearning.ai, Coursera.
- Generative AI with Large Language Models by DeepLearning.ai, Coursera.
- Game Theory by Stanford Online, Coursera.
- Heterogeneous Parallel Programming (CUDA), Coursera.
- Mining Massive Data Sets, Stanford Online.
- Data Scientist with Python, Data Scientist with R by Data Camp.

Scholastic Achievements

- GRE: **1370**/1600 {Quant: 800/800, Verbal: 570/800, AWA: 4/6} in Sep 2008.
- TOEFL: **113**/120 Read: 30/30, Listen: 30/30, Speak: 24/30, Write: 29/30 in Sep 2008.
- All India Ranks **773** in IIT-JEE 2003 and **652** in AIEEE 2003.
- State Ranks **810** in EAMCET 2003 and **19** in AIEEE B-Arch. 2003.

Co-curricular Activities

- Coordinator, Student's Council, IISc. 2011 - 2012.
- Creative Contributor, Telugu Samithi, IISc. 2010 - 2011.
- Associate, Janaagraha Center for Citizenship and Democracy. 2008 - 2009.
- Coordinator, National Service Scheme, IIT Madras. 2004 - 2007.
- Coordinator, MECHANICA, IIT Madras. 2006.
- Contributor, Campus Magazines, IIT Madras. 2005 - 2006.